Sohang Chopra

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Education

Indian Institute of Technology, Madras
Master of Technology in Industrial Artificial Intelligence — *Sept 2025 – Present*

Dayalbagh Educational Institute, Agra
Bachelor of Technology in Electrical Engineering, Minor in Computer Science — *July 2018 – May 2022*

Technical Skills

Languages: Python, C++, Rust, Bash, JavaScript, SQL
Skills & Concepts: Object Oriented Programming, Git, Open Source, GDB debugger, Operating System concepts (multi-threading, Mutexes, Semaphore, IPC, RTOS), Linux, Arduino, AWS, Docker, Machine Learning, Generative AI (LLMs), Full Stack Web, Github Actions CI, Web Scraping

Experience

Software Engineer — September 2024 – Present
HCL Technology — *Remote (Noida)*

- Consitently attained "Distinguished Performance" appraisal rating.
- Part of a team making a Python package to prune Tensorflow Vision models by 50 % so they run on resource-constrained chips.
- Worked on CI/CD Github Actions that automatically check code formatting & test it.
- Refined usability by improving error tracebacks and documentation, and automated testing & validation to minimize bugs.

Software Engineer — January 2022 – October 2024
MFilterIt — *Noida*

- Fine-tuned **Natural Language Processing** models with **Keras** and **PyTorch**, specialising in text detection and generation tasks.
- Implemented content monitoring and data collection tools using **Selenium**, **AWS Lambda** and **Pandas** to identify and analyse discrepancies in client web and mobile platforms.

Software Engineer (Intern) — April — September 2021
Arcon Cybersecurity — *Mumbai (Remote)*

- Made a flexible **GUI** Interface using **WPF Framework (C#)** to create automated web tasks through a flow chart.
- Enabled easy execution of created tasks multiple times using **Selenium**, a web automation library.

Projects & Open Source Contributions

IForth (Jupyter kernel for Forth) | *Python, Forth, Docker, Open Source*

- Revitalised the abandoned IForth **Open Source** repository by adding code prettification, shell command support, and stability enhancements, including Jupyter kernel termination management. Also added Dockerfile and published as Python package for easy installation.

Open Source Merged PRs | *JavaScript, Python, C, Rust*

- *C*: 2048.c: Added –help, –version CLI arguments
- *Rust*: Macrokata: Fix compile error in an exercise.
- *Python*: Added type hint stubs for Flask-SocketIO library to allow static type checking.
- *JavaScript*: ngn/apl: Made web app installable as a Progressive Web App (PWA) so it can be run offline on mobile or web.

JSON Parser from scratch | *Haskell*

- A JSON Parser implemented using parser combinators with Parsec library in Haskell (functional programming).

Embedded Morse Code Generator | *C++, Arduino*

- Developed a C-based Arduino application that converts user text input to Morse Code output via LEDs.

Electrical Charge Simulator | *C++, SFML*

- Simulated interaction of 100s of randomly charged particles in space.
- Studied many interesting phenomena (such as slingshot due to electromagnetic field).